Grassroots Networking

NAT Detection and Hole-Punching Survey

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1 Introduction

Grassroots Networking (GN) is one application-layer overlay over two transport underlays: a Bluetooth Low Energy (BLE) mesh for devices in radio range, and, when a wide-area network is available, the Internet, carrying UDP datagrams between trusted peers. The Internet underlay assumes that NAT traversal works. We have never measured whether this works for our users. Every constant in the transport, from the two-second punch to the ten-second connection timeout, was chosen by intuition. Public NAT measurements are dated and come from desktop and server vantage points [1, 2]. The hardest case is a smartphone behind a carrier-grade NAT (CGNAT). Such a phone cannot see its own public address, cannot accept unsolicited inbound traffic, and may get a new external port per destination, which defeats hole punching. Only code on smartphones with commercial SIMs can observe this.

The second problem is how a device learns whether it is reachable. libp2p's autoNAT asks other nodes to dial a peer back, meaning to attempt a connection to it, and sees whether the attempt arrives [5]. In practice it depends on a few well-known dial-back nodes, which is the privileged infrastructure a grassroots system is meant to avoid. GN already has a graph of mutually authenticated friends. We ask whether friends can serve as the dial-back nodes, which we call *witnesses*, and what accuracy we lose by replacing designated servers with them. The project designs and validates such an algorithm and runs it on a testbed we control. It compares the result against a reference: a server that implements RFC 5780, which needs two public addresses and two ports. We run one ourselves and call it the *oracle*. Finally, it measures real networks with our own smartphones.

2 Where the transport stands

The transport does not classify NATs. It binds one UDP socket per address family (IPv4, IPv6) on an OS-assigned port, and advertises the externally discovered address with its own local port, never the port the NAT assigned (`public_address_discovery.dart:97`). A device classifies itself with one true/false flag, `isWellConnected`, which tests whether the advertised address looks publicly routable. A phone behind a CGNAT advertises the carrier's public address, so it passes, and the pre-connect hole punch is skipped for exactly the peers that need it (`grassroots_network.dart:3369`). The punch itself is open-loop: 36 bytes sent for two seconds to one guessed address and port, with no feedback on whether it worked. A *reflection* is a peer telling a device which address and port it saw the device at. Reflections from several peers overwrite each other, so the disagreement between them, which is the raw material of a NAT test, is lost. Punch outcomes carry no timestamps and no failure reason, so a failed dial looks the same as an offline peer.

3 Research questions

RQ1: Can friends replace designated dial-back servers? We specify an algorithm in which friends act as witnesses, and measure how well its verdicts agree with the oracle's.

RQ2: What does decentralization cost? A dial-back can be lost for reasons unrelated to the NAT, so we ask how many witnesses and repetitions a confident verdict needs, and what it means when witnesses disagree.

RQ3: Which NAT behaviors occur on the networks our users inhabit, and how common is each?

We follow RFC 4787 and RFC 5780 [3, 4] and measure four properties instead of one NAT type, per address family:

- *Mapping behavior*: which external port the NAT assigns to outbound traffic. It is endpoint-independent if one port serves every destination, address-dependent if each destination address gets its own, and address-and-port-dependent if each destination address and port pair does.
- *Filtering behavior*: which inbound senders the NAT lets through an existing mapping. Same three levels: anyone, only addresses we sent to, or only the exact address and port we sent to.
- *Port preservation*: whether the external port equals our own local port.
- *Mapping lifetime*: how long an idle mapping survives.

autoNAT reports none of these [5], so the classifier is ours to write.

4 Approach

The *probe* is the device being measured. Its code lives in `lib/src/transport/` and sends from the transport's own socket, since a second socket gets a different mapping and measures a different path. Every witness is a friend, and the probe addresses each one directly. Friends are the only witnesses because GN signaling is friend-only, and a request that makes a third party send a packet is an amplification vector (a way to aim traffic at a victim), which autoNAT v1 refused to create. A friend does two things: it reflects, and it *dials back*, meaning it sends the probe a packet from a source address and port that it names in advance. Each dial-back must carry a *nonce*, a random number the probe chose for that request. A packet without the right nonce is ignored, so the report cannot be faked and an injected packet is rejected.

The procedure has two rounds. In round one each friend reflects. If every friend saw the same address and port, mapping is endpoint-independent. If friends at different addresses saw different ports but one friend, reached from two of its ports, reports the same port, it is address-dependent. If that friend reports different ports, it is address-and-port-dependent. In round two the probe requests three dial-backs: D_1 from an endpoint it has sent to, D_2 from the same address on a fresh port, and D_3 from an address it never sent to.

Arrives	**Filtering**
D_3	endpoint-independent
D_2, not D_3	address-dependent
D_1, not D_2	address-and-port-dependent
none	inconclusive

Three friends in the right positions, meaning two at distinct addresses and one that can send from two ports, separate every class. More friends only add confidence against loss. A dial-back that arrives is proof. One that does not is ambiguous, since it is equally consistent with filtering and with a lost packet, so the probe retries and asks k different witnesses before it records a negative. Disagreement between witnesses is kept and classified, never voted away. We assume friends are honest: that a friend which reports a dial-back sent it. Witnesses that lie are out of scope.

The main threat to validity is *contamination*. Friends are the peers the device already sends to, so a dial-back from one passes through a mapping the probe opened itself, and the NAT looks more open than it is. The *guard* prevents this. The probe logs every destination it sends to from the measured socket, and the guard discards any dial-back whose source is in that log. The strict filtering test therefore needs a friend the probe has no link to, and never sent a UDP packet to. The probe cannot ask such a friend over UDP, since that packet would open the very mapping under test. The request reaches it over BLE, or through a mutual well-connected friend that passes it on. The probe sends nothing to the friend's address, and only the friend's dial-back arrives. Repeating one arm with the guard switched off measures the bias.

To estimate how common each behavior is, we group networks into *strata*: networks that share the same NAT configuration, such as one carrier with one APN (access point name, the gateway configuration a phone uses to reach the carrier) and one PDN type (IPv4, IPv6 or both). Each cellular stratum is measured on at least two SIMs and on both Android and iOS. Residential networks are grouped per ISP and whether it uses CGNAT. Strata we cannot measure are carried as a residual. Prevalence is the sum of the strata weighted by published subscriber counts, reported with a sensitivity range and not a confidence interval. This belongs to the follow-on thesis, and this semester produces the per-stratum measurements it needs.

5 Experiments

Bench. The bench is a set of NAT configurations whose correct labels we know because we built them. We stand up the oracle and at least eight configurations. Linux netfilter, the kernel's packet filter, gives endpoint-independent mapping with its default masquerade rule and address-and-port-dependent mapping with the `-random-fully` flag, which picks a fresh port per flow. Address-dependent mapping has no flag, so we write per-destination source NAT in nftables, netfilter's rule language. The bench also spans the three filtering levels and a double NAT, two NATs in series, to imitate a CGNAT. Each configuration is measured by the oracle and by the algorithm with three witnesses in the same window. We record both labels, time to verdict and bytes sent, then repeat the filtering test with the guard off. The result is a confusion matrix (the constructed label against the label returned), the agreement with the oracle, and the contamination bias.

Witness count. On the bench we run witnesses at distinct addresses and ports and inject packet loss on the return path. We sweep k and the retry policy against the loss rate and record correct, inconclusive and wrong verdicts. The result is the k and retry policy at which a verdict can be trusted, and what each pattern of disagreement indicates.

Field. Our own smartphones with prepaid SIMs are moved across the networks we can reach, each measured on at least two days. There are no known labels there, so we report agreement with the oracle, not accuracy, and categorize every disagreement.

6 Plan

This is a praktikum, a one-semester supervised project of about 300 hours. The deliverables are the algorithm's specification with a reference implementation, the dataset, and a technical report.

Weeks	Work	Exit criterion
1–3	Onboarding. A STUN client in Dart, sending from the transport's own socket. STUN is the protocol by which a device asks a public server which address it appears at.	Public address obtained over that socket. Go/no-go: if the socket cannot be shared, we continue on a separate one and report that the mapping may differ.
4–6	Oracle and bench.	One command emits the confusion matrix, reproducible from a clean checkout.
7–10	Witness protocol, nonce, send log and guard, retry policy.	Labels recovered on the bench with three witnesses; contamination bias measured.
11–13	Field measurement.	All four properties per network, disagreements categorized, unreached networks listed.
14–15	Write-up.	Report, and one script that regenerates every figure.

Hosting. The oracle needs two public addresses per address family, each reachable on two UDP ports, and must receive unsolicited inbound UDP. A border firewall that drops such packets makes the host useless, so in week 1 we establish whether the university can grant an exception. The alternative is a commercial virtual machine with a second address. Witnesses need distinct addresses, not capacity, so several small hosts in different networks. A campus network is a good place to stand outside a NAT and a poor subject to measure.

Emulation. The bench runs in network namespaces, isolated virtual networks on one always-on Linux machine, driven by Mininet, a tool that builds them, or by plain `ip netns`. Mininet provides hosts, links and per-link loss, but not the NAT behaviors; those are still written by hand. An emulated NAT is only the outermost one if the host itself is not behind a NAT, and Docker's default bridge (its virtual switch) masquerades, so masquerading is switched off on it. A multi-node variant runs headless transport nodes, meaning the transport built without the Flutter user interface, each behind a NAT of a different class, befriending one another. Every ordered pair of classes can then be tested, and switching nodes off controls how many friends are online. It needs two code changes: the Flutter imports in seven files under `lib/src/transport` and `lib/src/signaling` have to go, and the function `isGloballyRoutableIPv4`, which tests whether an address is public, has to be replaceable, because it rejects every range an emulated network could use. Emulation reproduces a class of NAT, not a carrier, so the field tier stays on smartphones.

Excluded. Embedding autoNAT in the app, the prevalence estimate, and a rule that decides at dial time whether to punch, request help or abort. A manual check against public autoNAT nodes is reachability only, on Wi-Fi only. It characterizes the network and not our socket, and tethering on cellular would add a translation layer.

Success means the work is done properly and reported honestly. If the witness approach agrees poorly with the oracle, that is a finding: friends are not a sufficient substitute for designated servers.

7 Prerequisites and open decisions

Basic mobile development, basic networking, and statistics. Dart is learned in the first weeks. The student needs a Linux host, two public addresses, an Android and an iOS smartphone, and any

SIMs the measurement covers. Four decisions are taken before the work begins:

- Whether the probe sends from the transport's own socket. The week-3 checkpoint answers it. If not, a separate socket is used and the mapping caveat is reported.
- Who hosts the oracle and the witnesses.
- Whether to open IPv4 on the deployed rendezvous anchor, the always-on server that introduces peers to each other. Its rule admits `udp:9516` from `::/0` only, which is IPv6 only and excludes the IPv4-only CGNAT population we study.
- The SIM budget. One SIM covers one combination of carrier, APN and plan.

References

[1] C. Jennings. NAT Classification Test Results. IETF Internet-Draft, 2005.

[2] S. Guha and P. Francis. Characterization and Measurement of TCP Traversal through NATs and Firewalls. In *Proc. ACM IMC*, 2005.

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[4] D. MacDonald and B. Lowekamp. NAT Behavior Discovery Using Session Traversal Utilities for NAT (STUN). RFC 5780, 2010.

[5] Protocol Labs. libp2p AutoNAT v1 and v2 specifications. `https://github.com/libp2p/specs/tree/master/autonat`.

[6] D. Seemueller et al. A Longitudinal Measurement of Hole Punching in the IPFS Network. arXiv:2510.27500, 2025.